

Manual

Editing of Orders/Work Plans (MOC)

BDE-BAA 8.1

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1 Übersicht Bearbeiten Aufträge / Arbeitspläne

Purpose

The function package “editing orders/work plans” provides an extensive range of functions for

- creating or modifying orders with operations in the system
- creating and editing work plans in the system
- creating orders based on work plans

Implementation considerations

You use the function package if

- You have no interface to an ERP system from where orders are retrieved.
- You have an interface to an ERP system from where orders are retrieved and you
 - would like to modify the transferred orders/operations in MES,
 - want to create and manage your own orders in addition to the orders transferred from the ERP system.
- For activities that are performed over and over again, you want to use a work plan as a basis for creating new orders.

Features

- Order management
 - Function for creating production, overhead cost or rework orders
 - Assigning 1-n operations to an order
 - Assigning material components, production resources and documents to operations
 - Assigning long texts to orders and operations
 - Option to modify and correct order backlog data supplied by ERP/PPS systems
- Work plan management
 - Creating copy masters of work plans used to generate production, overhead cost and rework orders
 - Function used to generate production, overhead cost and rework orders from work plans
- Generating orders
 - Creating production, overhead cost or rework orders based on work plans
 - Creating production, overhead cost or rework orders by copying already existing orders

2 Objekt Auftrag

Definition

An order in MES is generally a request to production to carry out a certain action accounting for a variety of work steps. A distinction is generally made between a production order and so-called overhead cost orders.

1. 1. The production order (production) defines:

- The material/ article to be produced
- The quantity to be produced (batch size)
- The earliest and latest start and finishing dates

2. 2. The overhead cost order (service) defines:

- A particular activity
- The calculation reference

The order in MES primarily consists of

- The order header information
- The operation data

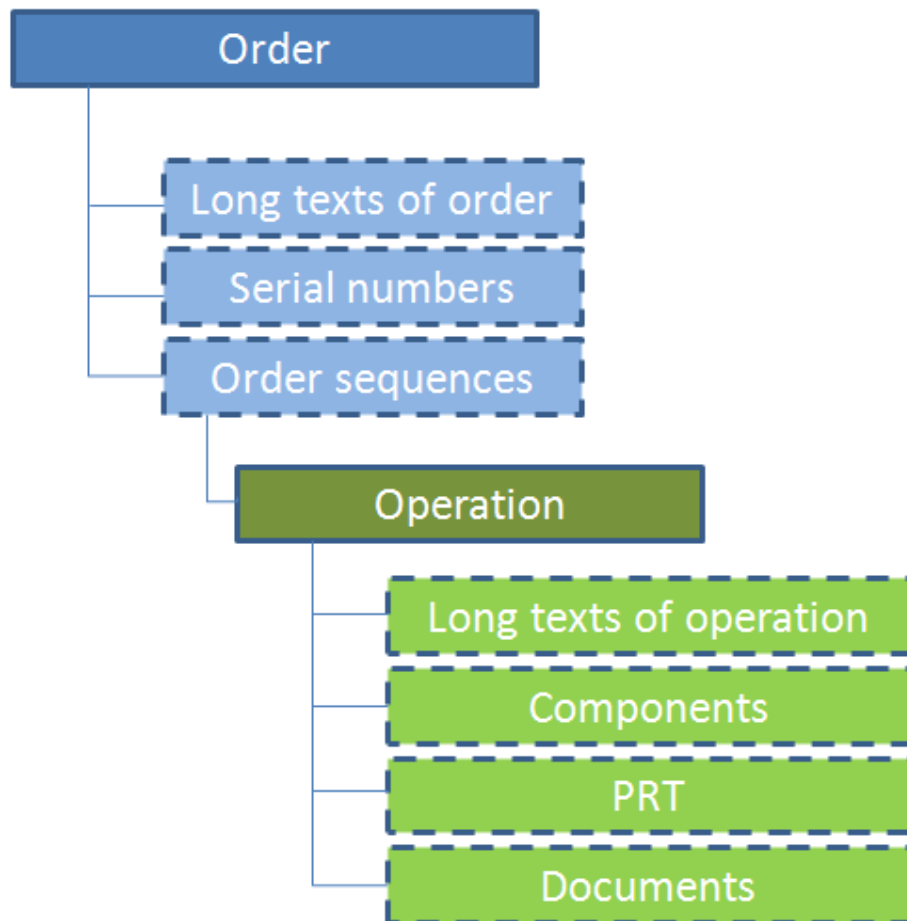
Usage

In MES the order (header) information is used to complete the shop floor papers and to manage the data, which is the same for any operation/process of a given order.

All activities that a person carries out on a machine/work station are order and/ or operation related. The posting of the order and operation answers the question what is being done and/or what activity is being carried out.

Structure

Every order is identified by a unique ID or order number. This is either provided and administered by an upstream system (generally ERP systems) or by the MES system itself. The object Order is structured as follows:



Integration

The order includes n operations that are to be carried out. The order thus produces a certain material or final product with a certain type of material.

3 Objekt Arbeitsgang

Definition

The operation or procedure is one step within a work flow during which a manufactured quantity of an order's article is produced.

In addition to the operation number, the information listed below is needed to identify the operation:

- A written description of the work that needs to be performed
- The required workplace or the required group of identical workplaces
- In some cases, any other resources needed (e.g. tools, drawings, NC programs)
- Time needed to carry out the work (e.g. setup time, processing time)
- Target quantity (batch size)

If different machines or groups are required for production, this is what we refer to as multilevel production. Therefore, multilevel production includes several machine-related operations, which normally run one after the next. The number of operations needed for an order is not limited.

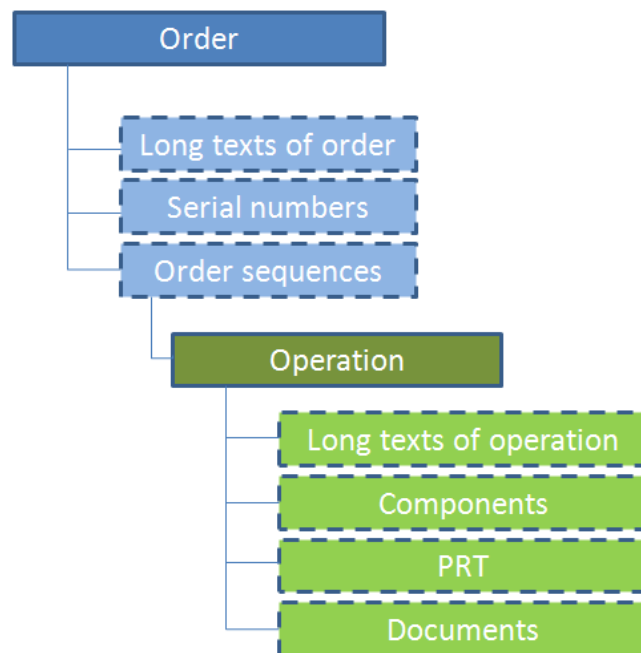
Terms used synonymously for the term operation are: procedure or order sequence/maintenance sequence (AFO). Oftentimes, the term order itself is also used synonymously.

Usage

All activities that a person carries out on a machine/work station are order and/or operation related. The posting of the order and operation answers the question what is being done and/or what activity is being carried out.

Structure

Each operation can be identified by the relevant combination of the unique order number and the sequence and operation number. This is either provided and administered by an upstream system (generally ERP system) or by the MES system itself. The object "operation" is subordinate to the object "order" and "sequence" and is structured as follows:



Please note: the object order sequence is only used if specifically requested.

Integration

The operation outputs a material with a specific material type. The operation also includes as additional information the bill of materials or rather the component list showing the materials that are needed or that are relevant in manufacturing the article. The same applies to the range of different production resources (e.g. tools) itemized in the production resources and tools list.

4 Edit Orders

Summary

Menu	Order Management → Order Management → Edit Orders
Transaction code	edor
Function authorization	edor

Usage

This document provides a description of how orders can be created and edited in MOC.

Integration

Typical applications that require orders to be edited include:

- Creating overhead costs orders
- Creating orders if no ERP system is available
- Correcting order stock data

Also described in this document is the order structure, i.e. the fields relating to the order header.

Prerequisite

The following configurations must exist

- Order types

Selection criteria

The application provides the following selection criteria:

Order

This selection criterion references the order number. The selected order is displayed. There is an option to enter wild cards.

Order type

This selection criterion references the order type. All orders with the selected order type are displayed.

Article

This selection criterion references the article in the order header. All orders with the selected article are displayed. There is an option to use wild cards.

Sales order

This selection criterion relates to the sales order defined in the order header. All orders are displayed assigned to the sales order. There is an option to use wild cards.

Project number:

This selection criterion relates to the project number defined in the order header. All orders of the selected project number are displayed. There is an option to use wild cards.

Planned order

This selection criterion relates to the planned order defined in the order header. All orders of the selected planned order are displayed. There is an option to use wild cards.

Customer designation

This selection criterion relates to the customer name defined in the order header. All orders of the selected customer designation are displayed. There is an option to use wild cards.

Checking the responsibility area

During the selection, the responsibility area defined for the order is checked.

Field descriptions

The separate fields in the order header are described [here](#). The sequence described there may deviate from the sequence in the editing dialogs.

Toolbar

Function authorization: or.generate

Starting the "generate order" dialog

Please note: If orders are generated by this function, the [work plan determination](#) function is used. To generate an order from a specific work plan, please use the "generate order" function in the [Work plans - generate order](#) application.

**Edit long texts of orders**

Function authorization: edortx

Link to function: [Edit long texts of orders](#)

**Edit order sequences**

Function authorization: edseq

Link to function: [Edit order sequences](#)

**Edit operations**

Function authorization: edop

Link to function: [Edit operations](#)

**Order information**

Function authorization: orin

Link to function: [Order information](#)

**Order overview**

Function authorization: orov

Link to function: [Order overview](#)

5 Edit Long Texts of Orders

Summary

Menu	Order Management → Order Management → Edit Long Texts of Orders
Transaction code	edortx
Function authorization	edortx

Usage

By applying the function “edit long texts of orders”, order-related additional texts can be displayed or edited. You use this function if:

- You would like long texts belonging to the order header to be visible and available in the administrative client while processing the order.
- You are using the MES Development Suite Label Designer component and the data you entered is to be printed on labels.

Keep in mind that for each order you use a maximum of one long text.

Integration

Long texts can also be transferred via the info interface (record type "AI"). Additional information about the interface can be found in the respective interface document.



Only long texts relating to the operation are displayed at the terminal.

Prerequisite

The corresponding order must already be defined.

Long texts included in the online data area may generally be edited, i.e. irrespective of the order status (added, modified or deleted).

Selection criteria

The application provides the following selection criteria:

Order

The long text for a specific order can be selected by entering the order number.

Field descriptions

The fields for long texts of orders are described [here](#)

Editing functions

To create a new operation or to edit one, you use the icons provided.

The long text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

Toolbar



Edit orders

Function authorization: edor

For the currently selected data record, this will call up the application [Edit orders](#).

6 Edit Order Sequences

Summary

Menu	Order Management → Order Management → Edit order sequences
Transaction code	edseq
Function authorization	edseq

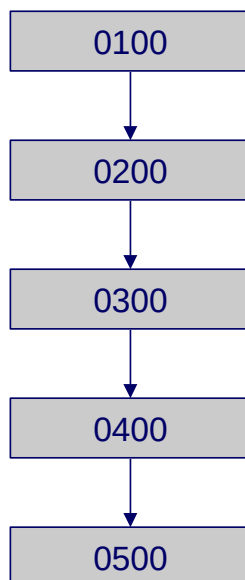
Usage

Operations within an order are grouped into sequences in order to create a summary of them. Production uses this information as an orientation tool to process each operation. Within the sequence, the operations are processed in sequence one at a time. By linking several sequences within the order, network-type structures can be illustrated.

You also have the option to use parallel or alternative order sequences. The following sequence types are supported:

Standard sequence

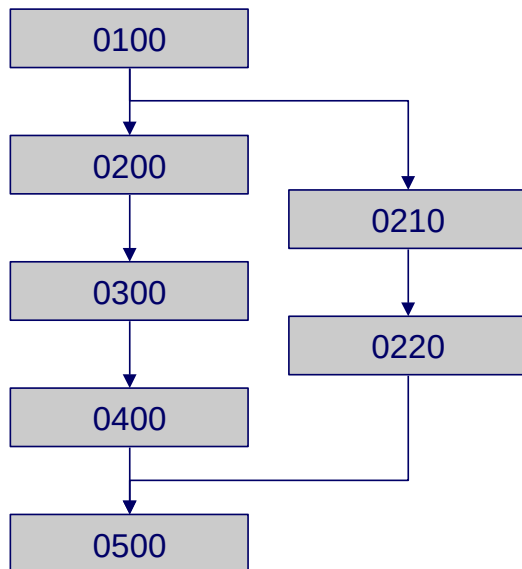
The standard sequence is available by default and describes the first sequence of the order.



For a purely sequential order, only the standard sequence is required. If certain operations are to be processed in parallel or alternatively to the standard sequence, they must be grouped in relevant sequences. Thus, parallel and alternative sequences can branch off of a single standard sequence.

Parallel sequences

A parallel sequence runs parallel to a partial sequence of the standard sequence. It is used, for example, when certain processes are to run at the same time (in parallel). This may be the case, for example, in the processing industry.



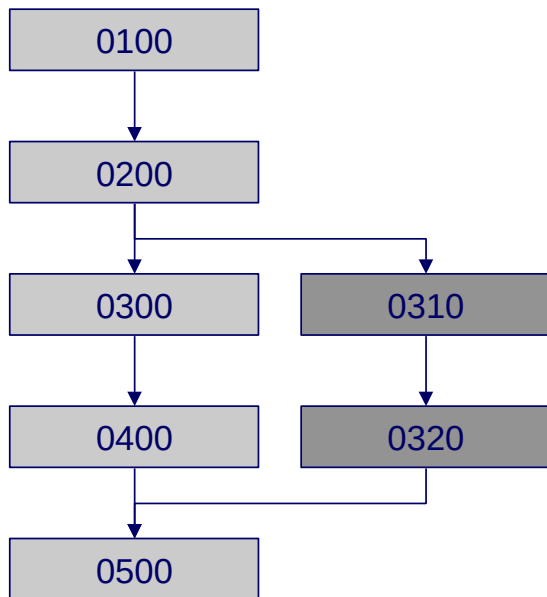
This partial sequence is defined by the branch operation and the return operation of this particular reference sequence. As such, the start of the parallel sequence is equal to the start of the branch operation in the reference sequence and the end is equal to the end of the reference sequence's return operation.

Alternative sequences

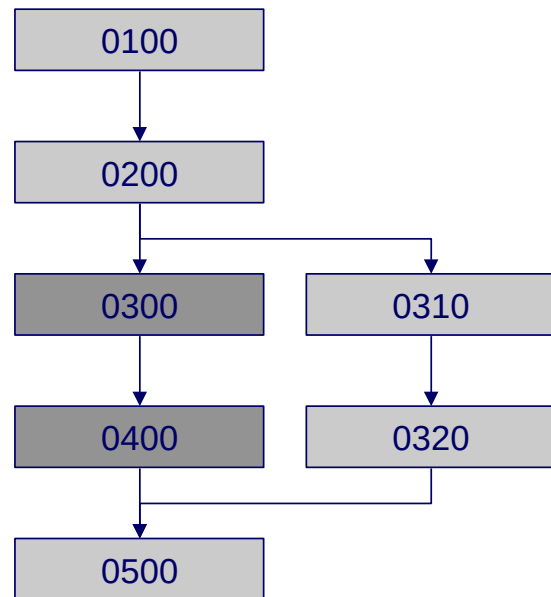
An alternative sequence describes one or more operations, which can be used alternatively to a partial sequence of the standard sequence. It is used, for example, if the production process varies for certain batch sizes.

Alternative sequences each have one active sequence that is relevant for processing.

Order with an inactive, alternative sequence



Order with an active, alternative sequence



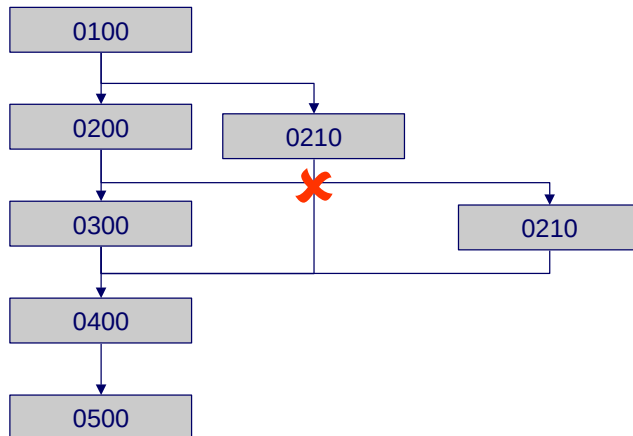
Inactive sequences are not considered either in scheduling or in detailed planning; they can neither be posted.

It is possible to activate an alternative sequence at the HYDRA console if certain conditions are met. This occurs interactively.

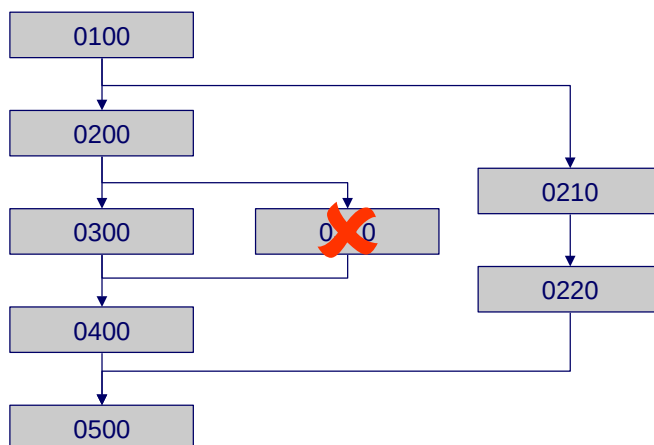
General

- Each operation is assigned to exactly one sequence.
- The relationships that are created as a result between the operations and between the sequences are stored by HYDRA in an internal relationship table.
- Parallel sequences always run parallel to the standard sequence.
- Alternative sequences always run alternately to the standard sequence.

- Operation sequences of different sequences may not overlap.



- With respect to a partial sequence of the standard sequence, which is restricted by a branch OP and a return OP of a parallel or alternative sequence, there may be no parallel or alternative sequence with a branch operation and/or return operation within it.



Deleting sequences

- A sequence may only be deleted if there is no operation that is assigned to this sequence.
- As a rule, a standard sequence cannot be deleted.

Creating and deleting operations

- When an operation is created or deleted, HYDRA automatically updates the order network for this order. The order network documents the relationships between operations and this information is used for planning in HYDRA shop floor scheduling (HLS) as well as for processing/ posting.

Copying an order

- If an order is copied, the new order is available in its "initial state". This means that any existing alternative sequences are generally inactive, even if they were previously active in the order that was copied.

Operation status

- It cannot be determined based on the operation status whether the operation is a part of the standard sequence's active alternative sequence or its inactive alternative sequence or rather its inactive partial sequence (as a result of activating an alternative sequence), because the operation status does not change during activation or deactivation.
- Operations of an inactive alternative sequence have the same initial status *prepared* when they are newly created, just as is the case for operations of active sequences.

Sequencing list

- Operations of the standard sequence's inactive alternative sequence or an inactive partial sequence (as a result of activating an alternative sequence) are not shown in the sequencing list. They can neither be logged on.

Merged operation

- Operations of the standard sequence's inactive alternative sequence or inactive partial sequence (as a result of activating an alternative sequence) cannot be a part of a merged operation.

Issues of note relating to parallel sequences

- Parallel sequence structures are accounted for in the planning algorithms in HYDRA shop floor scheduling (HLS)
- If the option "Checking status of preceding OP" is active (customized HYDRA feature), after merging several parallel sequences, an operation may not be logged on until all sequences or rather their last operations have been interrupted or have been finished (depending on the customized settings). Any overlapping is considered in the process.

- The smallest yield of the consolidated sequences, or rather of their last operations, is considered the send-ahead quantity of several parallel sequences. This is carried forward as the target quantity to the successors if the processing code provides for a target quantity update at the operation (customized HYDRA feature).

Integration

Please note the following with regard to displaying operations of alternative sequences in the MOC functions and evaluations/reports:

Operations/ operations logged on/ pool of orders

An operation of an inactive sequence ("inactive operation") can be recognized by the "Y" in the column *Control*.

If no operations of inactive sequences are to be displayed in the order overview, then all of the options except for the option "Y" must be set in the selection range in the *Control* selection field.

Order overview

In the *Progress* index tab, operations of both active as well as inactive sequences are displayed.

Order information

In the order information, operations of both active as well as inactive sequences are displayed. In order to be able to recognize inactive sequences as such, you use the column configurator in the operation table to have the *control* column displayed. In this column, operations of inactive sequences are listed with a "Y".

Requirement

In order to process sequences in HYDRA, the relevant license must have been issued. It is not possible to use DOS based terminals.

The following activities are required for use:

1. The sequence number length must have been set in the basic HYDRA settings

WARNING

You may only set the sequence number length during the initial HYDRA setup process and provided that no order backlog data exists in HYDRA. Any subsequent settings or changes will make the system act inconsistently.

2. Reactivating dynamic dialogs

As a result, the input fields at the Windows terminal will expand by the defined sequence number length.

Selection criteria

The application provides the following selection criteria:

Order

Enter the order number for the order with the sequences you would like to display.

Field descriptions

Order number

Order for which the sequence is defined. A sequence can only be set up for an order, if the order header already exists in HYDRA.

Sequence

Identification of the sequence within an order.

Please note: The standard sequence always has the sequence number 0.



If the "sequence" field is not shown in the editing dialog, the sequence number length is 0 in the basic parameter settings. Please contact MPDV.

Designation

Description of the sequence.

Sequence category

S = Standard sequence

There is only one standard sequence for every order; the standard sequence cannot be deleted.

P = Parallel sequence to the standard sequence

There can be several parallel sequences for each order

A = Alternative sequence to the standard sequence

There can be several alternative sequences for each order

Please note:

The sequence category cannot be edited after a sequence has been set up!

Active

The qualification "Active" is only relevant for alternative sequences:

J = Active

N = Not active

If a new alternative sequence is set up, it is set as *not active*.

For standard sequences and alternative sequences, this qualification is always set to *active*.

Orientation

If there are several parallel sequences, the lead times generally vary in length. This creates time buffers in the sequences. The orientation function controls whether these buffers are at the beginning or the end of the sequence. The following options are available:

F = Earliest due date

If the sequence is set for the earliest date, the buffer will be at the end of the sequence.

S = Latest due date

If the sequence is set for the latest date, the buffer will be at the beginning of the sequence.

N = Not relevant; this is the case for standard sequences and alternative sequences.

If there are several parallel sequences for a given standard sequence, the orientation of the standard sequence is used for all segments of the standard sequence for which parallel sequences exist.

Version

Change number/ version; for information purposes only.

Branch operation

Operation number of a standard sequence operation,

- *before* which a parallel sequence should branch off, or
- *from* which an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch operation of the standard sequence must already exist in the order backlog. When a sequence is handed over via an interface, a valid operation number also must be handed off (there is no validation check).

Return operation

Operation number of a standard sequence operation,

- *after* which a parallel sequence should return, or
- *up to* which an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch-off operation of the standard sequence must already exist in the orders backlog. When a sequence is handed over via an interface, a valid operation number also must be handed off (there is no validation check).

Reference Sequence

The reference sequence determines the sequence in the order that the reference operations (branch and return) refer to. This is *always* the standard sequence (sequence number 0).

This is a mandatory field for parallel and alternative sequences. The standard sequence must already exist.

For a standard sequence, this field must remain empty.

Toolbar



Activate

Activate an alternative sequence



Deactivate

Deactivate an alternative sequence



Edit orders

Calls up the application [Edit orders](#).

7 Edit Operations

Summary

	→→	
Menu	Order Management → Order Management → Edit Operations	
Transaction code	edop	
Function authorization	edop	

Usage

The term work procedure or operation is used to refer to a work sequence within a work system geared towards the completion of a work task, during which an order's quantity unit is created.¹

You use this function to add new operations to an order or to modify data in existing operations.

Integration

Operations are planned in planning functions (e.g. HLS) and are logged on to shop floor terminals; their purpose is to facilitate status tracking and to record quantities and activities, which are then typically uploaded to upper-level systems.

Requirement

The following requirements must be met when adding a new operation:

- The superior order must already have been created.
- When making use of order sequences (project-specific; depending on the license), the sequence that the operation is to be associated with must already exist.
- The workplace/ machine or group on which the operation is to be planned has already been created in the system.

You have been assigned to the responsibility area and are authorized to display the data.

¹ REFA Association for work studies and business organization e.V. (publisher): Methodology of Organizing a Business: Encyclopedia of Business Organization. Munich: Carl-Hanser, 1993. - ISBN 3-446-17523-7. Page 195.

Selection criteria

The application provides the following selection criteria:

Order

Enter the order number for the order with the operations you would like to display. You can also run a search using wildcards.

Operation

You have the option to enter the operation number for the operation that you would like to display or edit. You can also run a search using wildcards.

Sequence

If your system is set up to use sequences (depending on the license), you can enter the sequence number here. In this case the operations are considered that are assigned to the sequence numbers you entered. If your system is not set up for sequences, please leave this field empty.

Display split OPs

If you use the function provided for splitting operations (depending on the license), you can use this option to define whether in addition to the split-master-operations, their splits should be displayed as well.

Checking the responsibility area

During the selection, the responsibility area defined for the operation is checked.

Field descriptions

The fields for the operation are described [here](#)



Only selected data is available in the table:

- Order
- Sequence
- Operation
- Split
- Processing code
- Locked
- Fixed

- Group
- Workplace
- Control

Editing functions

To create a new operation or to edit one, you use the icons provided.

If a responsibility area is defined at the order, data maintenance (editing) depends on how the options display, insert, modify and delete are set up in the responsibility area configuration or in the responsibility profile.

Toolbar



Edit orders

Function authorization: edor.*

Calls up the application [Edit orders](#) for the selected order.



Edit order sequences

Function authorization: edseq.*

Calls up the application [Edit order sequences](#) for the selected order.



Edit operation long texts

Function authorization: edoptx.*

Calls up the application [Edit long texts of operations](#).



Edit components

Function authorization: edopcomp.*

Calls up the application [Edit components](#).



Edit production resources and tools

Function authorization: edopres.*

Calls up the application [Edit production resources and tools](#).



Order information

Function authorization: orin

Calls up the application [Order information](#) for the selected order.



Modify operation status

Function authorization: op.statchg

Function to [modify the operation status](#)



Lock

Function authorization: op.lock

The button [block operation](#) locks one or several selected operations.



Unlock

Function authorization: op.unlock

The button [unlock operation](#) unlocks one or several selected operations.



Splitting an operation

Function authorization: op.split

Calls up the function to split the operation. Additional information can be found in the respective [document](#).



Dissolve split OP

Function authorization: op.splitdissolve

Cancels the split operation. Additional information can be found in the respective [document](#).

Adding an operation

Transferring order header data

The following data is transferred from the order header in the operation when a new operation is created:

- Order type
- Base quantity unit
- Article if the article number is not explicitly defined at the operation.
- Article designation if the article designation is not explicitly defined at the operation.
- Material type if it is not explicitly defined at the operation.

- Customer designation
- Priority, if the priority control is set to *order-related* at the order type..



Any priority that may have been entered will be ignored!

Transferring default data

Default data is taken from a template or from the processing code, if one exists, and transferred to the operation when an operation is created. The data is transferred in the following order:

- Values are transferred from the template (if one exists)
 - When adding a new operation (manually or by interface), all values are taken over that can be edited in the template but that are not entered manually (explicitly) or by the interface.
- Values are transferred from the *processing code* (if one exists); doing so will overwrite any values set in the template. The following values are transferred from the processing code into the operation:
 - Underdelivery
 - Reaction to underdelivery
 - Overdelivery
 - Reaction to overdelivery
 - External processing
 - Recordable
 - Can be logged on multiple times
 - May be split
 - Serial number obligation
 - Batch management requirement
 - Target quantity update*

- Sequencing list* is no longer evaluated for display in the sequencing list; instead, the system accesses the separate configuration tables directly.

Please note: Values marked with * are not displayed at the operation and therefore they also cannot be changed.

- Transfer of the values transmitted explicitly (either entered manually or transmitted via PPS interface); doing so will ignore any values that were previously transferred from the template or the processing code and are thus now overwritten.

Target quantity comparison

If the target quantity comparison at the preceding operation is active, when an operation is inserted, any target quantity that was entered is ignored and is assumed from the preceding OP.

Determining transport time

When customizing HYDRA a transport matrix can be defined that is then used to determine the transport time between two operations. This transport time is then considered during lead time scheduling.

When a new order or operation is created, the transport time is calculated using this matrix and then transferred to the operation. Any change to the transport matrix later will have no effect on already existing operations.

If, at the time operations are transferred at the interface, a transport time is set greater than zero for the ERP system, it is transferred into the database. Otherwise, the time is calculated from the transport matrix.

Irrespective of the master data, the values in the operation can be modified explicitly. What needs to be considered here is that any values modified explicitly are then overwritten when an operation is re-planned to another machine group.

→

→→→Setting of planned start data (to be used for the [sequencing list at the terminal](#))

When a new data record is added, the system tries to determine planned start dates and to use them as default values for sorting the sequencing list. This process is based on the following logic:

- It is checked whether or not the planned start date and the planned start time are empty.
 - If this is the case:
 - The planned start date is taken from the earliest start date (date)
 - The planned start time is taken from the earliest start date (time)
- It is checked whether or not the planned end date and the planned end time are empty:

- If this is the case:
 - The planned end date is taken from the latest end date (date)
 - The planned end time is taken from the latest end date (time)
- For sorting of the sequencing list the planned start date and the planned start time are entered in separate fields that cannot be changed.

The used date fields, the corresponding BAPI acronyms and database fields can be found in the document dealing with the technical [background information on the sequencing list](#).

Editing an operation

Default data is taken from a template or from the processing code, if one exists, and transferred to the operation in the following order when an operation is modified.

- If a planned group was modified or the workplace resulting in a different group, then the following values will be transferred from the template, if one exists:
 - Wait time formula
 - Setup time formula
 - Processing time formula
 - Inspection time formula
 - Dismantling/teardown time formula
 - ←←←←Target cycle formula
 - Formula for the remaining run time formula
 - Formula for the second remaining run time formula
 - Max. synchronization time
 - Default value key
- If the planned group was modified, the system will moreover also update the value plan_werk (internally managed in the order backlog) in which the company for the modified [Group](#) is calculated and transferred.
- If the processing code was changed at the operation, these values will be transferred from the processing code (see above).

- The final step involves the transfer of the values transmitted explicitly (either entered manually or transmitted via ERP interface); doing so will ignore/ overwrite any values that were previously transferred from the template or the processing code.

If the group for an operation is modified, the transport time will also be recalculated. In doing so, the transport time stored in the transport matrix will be assumed. Any previously defined transport time in the preceding OP will thus be ignored/ discarded.

Transferring order header data

When an operation is modified, only the following values from the order header will be transferred into the operations:

- Priority, if the priority control is set to *order-related* at the order type.



Any priority that may have been entered will be ignored!

- Customer designation

The base quantity unit will not be modified, because it is very unlikely that a change of this kind would ever happen in reality. The material type is not modified, because in MPL it may vary from one OP to the next OP.

General checks run when storing an operation

Checking for the existence of a workplace or group

If a workplace was entered, then the system also checks whether or not it exists. If so, then the group will be transferred into the operation that is assigned to the workplace. In any other case, the storing process will be interrupted with an error message.

If no workplace was defined, but instead only a group, then the system will check the validity of the group that was entered. That is, it will check if the group exists in the system. If this is not the case, the change will be rejected and an error message will be issued.

Checking priority management

If priority management was activated for the order type during the customizing process using the identifier with the same name and the priority control was set as order-related, then when an order is newly created manually or modified, the system checks whether the defined priority is permitted and, in the event of a violation of the maximum number, the action will be rejected.

If the order comes from the ERP interface and the maximum number is exceeded, the order will not be refused as a result of this validation check. In this case, however, the priority will be automatically set to 1.

Ability to modify an operation

In the standard version, the following OPs cannot be modified:

- OPs that are currently logged on (status with control indicator L) as well as
- OPs that are automatically interrupted (status with control indicator F).

Setting of planned start data (to be used in the [sequencing list at the terminal](#))

When an existing data record is changed, the system tries to determine planned start dates, to update them and to use them as default values for sorting the sequencing list. The basic logic depends on the planning function configuration in the master record of the operation's workplace:

- Planning function "N" – no planning
 - It is checked whether or not the planned start date and the planned start time are empty.
 - If this is the case:
 - The planned start date is taken from the earliest start date (date).
 - The planned start time is taken from the earliest start date (time).
 - It is checked whether or not the planned end date and the planned end time are empty:
 - If this is the case:
 - The planned end date is taken from the latest end date (date).
 - The planned end time is taken from the latest end date (time).
 - For sorting of the sequencing list, the planned start date and the planned start time are entered in separate fields that cannot be changed.
- Planning function "P" / "H" / "T" / "A":
 - If the planned start date is changed the planned start date and the planned start time will be entered in separate fields that cannot be changed.



If no workplace is defined for the operation, processing will be performed along the lines of the planning function "N".

The used date fields, the corresponding BAPI acronyms and database fields can be found in the document dealing with the technical [background information on the sequencing list](#).

Deleting operations

When an operation is deleted, the following points must be considered:

- In the standard version, an operation can only be deleted if the operation is not logged on (i.e. the OP is neither in the status "running" nor in the status "automatically interrupted").

- A split operation cannot be deleted. In this case, it must be canceled using the relevant splitting function.
- If a split master is deleted, this will also delete its split OPs.
- A merged operation cannot be deleted. In this case, you must cancel it.
- If the last operation for an order is manually deleted from the console, the order header itself will not be deleted. It must be deleted explicitly.

In the standard version, deleting an operation means that an item is physically deleted from the database. In the process, the following data is deleted:

- Order backlog
- Order status
- Assigned material components
- Assigned production resources and tools
- Assigned long texts
- Resource allocation for this operation in shop floor scheduling (HLS).

The log data is not automatically deleted if an operation is deleted. These are transferred into the long term table or are deleted from the database during cyclical archiving/ deletion runs.

8 Edit Long Texts of Operations

Summary

Menu	Order management → Order management → Edit long texts of operations
Transaction code	edtx
Function authorization	edoptx

Usage

Applying the function edit long texts of operations, operation-related additional texts can be displayed or edited. What should be considered in this regard is that only a maximum of one long text can be recorded/assigned to an operation at any one time.

Operation-related long texts can be displayed at the terminal.

Long texts can also be transferred via the HYDRA info interface (record type "AI"). Additional information about the interface can be found in the respective interface document.

Requirement

The corresponding operation must already be defined.

Long texts included in the online data area may generally be edited, irrespective of the operation status (added, modified or deleted).

Selection criteria

The application provides the following selection criteria:

MES order number

Entry of the combined order/ operation number. There is an option to use wild cards, for example in order to be able to display all an order's operation-related long texts. In this case, the order number must be entered, followed by *.

Field descriptions

MES order number

The operation's combined order/ operation number. This is a mandatory field.

Short Text

20-digit short text that is displayed in the table view. This is a mandatory field.

Long Text

The order's long text.

The long text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

Toolbar

Edit operations

Calling up the application: [Edit operations](#)

9 Edit Notes

Summary

Menu	Order Management → Order Management → Edit Notes
Transaction code	ednotes
Function authorization	edopnote

Usage

The function "edit notes" can be used to edit operation notes.

Requirement

The corresponding operation must already be defined.

Notes included in the online data area may generally be edited, i.e. irrespective of the operation status (added, modified or deleted).

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/ operation number.

Please note that the components are assigned by specific operations. This is why the entire key must be entered. By entering the order number followed by *, the system will list all components for an entire order.

Field descriptions

MES order number

Combined order/ operation number.

Short Text

Short text of the note

Long Text

Long text of the note

Display on terminal

Specifies whether or not this operation note is shown on the terminal.

Toolbar



Edit operations

Calling up the application [Edit operations](#).

10 Edit Components

Summary

Menu	Order management → Order management → Edit components
Transaction code	edcomp
Function authorization	edopcomp

Usage

The materials needed to produce an article are assigned to the operation as so-called (material) components. This function makes it possible to display or edit material components belonging to the operation.

Generally these components are transferred via an interface from an upper-level ERP system to HYDRA, because these are already defined at the operation there.

Requirement

The corresponding operation must be defined.

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/ operation number.



Please note that the components are assigned by specific operation. This is why the entire key must be entered. By entering the order number followed by *, the system will list all components for an entire order.

Field descriptions

MES order number

The combined order/ operation number for the operation that is to be assigned to the production resource or tool is entered here.

Material

Enter the material number for the material component here.

Designation

Here you can specify the name of the material.

Comment 1/ Comment 2

These are comment fields.

BOM item

In a BOM, the separate components of a product are referred to as items. The order of the BOM item is determined by the number that is entered here. As a result, the same material numbers can appear in the component list and the unique entry can be assigned to the correct component item.

If the production is roll-based, this is then the position of the component in the layer structure.

Keep in mind that the BOM item is an integral part of the identification key.

BOM level

A component can also have several levels. If applicable and known, please enter the BOM level here.

Please note: Postings can only be carried out on materials on a BOM level 0. If the system indicates a BOM level > 0, the component type (see next field) must as a rule be set to "I" (info component).

Component type

Component type Possible values:

M	Material component This component type must be defined in it. Other types can also be relevant here as a part of MPL or its roll-based MPL-RF solution.
I	Info component. Info components provide the ability to be shown in the component list without having to post them.
T	Carrier material (MPL-RF). A maximum of one input batch may be logged into the machine as carrier material (T) or additional material (Z) at any one time.
Z	Additional material as an alternative for carriers.
A	Scrap/ waste material

Material type

Type of material of the material component. In HYDRA, material specific processing is controlled by the material type.

Unless defined otherwise for a specific project, assign the material type SYSTEM here.

The material type must exist in HYDRA. If no material type has been entered, HYDRA will attempt to determine the material component (requirement: the assignment of material by material type has been entered). If no material type can be found, HYDRA will assign SYSTEM as the material type.

Please note: For info components (material type "I") you should define and assign a separate material type (e.g. INFO).

Consumption type

The following options are available to enter material components. How the separate options and their application are defined depends, among other things, on which of the HYDRA modules are used.

K = None

This option defines that no consumption is recorded for the material component. If this is the case, the material component will merely be displayed.

It is imperative that the info components (see above: component type) are set to this option.

D = Discrete

For this component, the system calculates consumption in reverse at the Windows terminal, which means the calculation is based on the last produced quantity and suggested in a posting dialog. Consumption is posted at the component level and the system generates a material flow (goods issue from production), which can be uploaded to the upper-level ERP system (requires the relevant interface for uploading goods movements).

Recording material consumption in this manner requires a custom configuration that can be performed by customizing HYDRA at the customer's location.

L = with batch reference (relevant when MPL/ TRT is used)

Choosing this option sets the system so the material component is logged in and out as a HYDRA batch. The consumption calculation for these material components (retrograde, at input batch logout) depends on the configuration of the material type that the material component is assigned to.

Requirements quantity/ unit

Planned total quantity for the component in the operation.

It is calculated automatically using the operation's target quantity (primary quantity unit) multiplied by the component's input quantity.

The requirement quantity is only shown in the table and in the detail panel.

Input quantity

Planned input quantity of the component per unit of the primary quantity of the operation.

MPL, consumption type D: Planned input quantity of the component per unit of the primary quantity of the operation.

Unit

Quantity unit for input quantity.

Input quantity in %/ upper tolerance limit/ lower tolerance limit

Reserved: Not edited. Should be set to 0.

Replaceable

If this identifier (=J) is set for this kind of component, you can choose to use a different material other than what was planned. However, only material of the same material type may be used.

Requirement to change output batch

An input batch change for a batch of this material requires a change to an output batch. What must be considered here:

☒, if type = T or Z

☐, if type = I or A

☒☐, if type = M

Otherwise: N

Upper-level component: BOM item/ BOM level

Reserved: Not edited.

Toolbar



Edit operations

Calling up the application [Edit operations](#).



Edit orders

Calling up the application [Edit orders](#).



Order information

Calling up the application [Order information](#).

11 Edit Production Resources and Tools

Summary

Menu	Order management → Order management → Edit production resources and tools
Transaction code	edres
Function authorization	edres

Usage

Resources can be defined for operations in the list of production resources and tools.



Further information on how to define workforce requirements via production resources and tools can be found in the document entitled [Definition of Workforce Requirement.pdf](#)

Requirement

The corresponding operation must already be defined.

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/ operation number.

Please note that the components are assigned by specific operations. This is why the entire key must be entered. By entering the order number followed by *, the system will list all components for an entire order.

Field descriptions

Order/ operation

Enter the order/ operation number for the operation that is to be assigned to the production resource or tool here.

Resource type

Resource type of the production resource or tool that is to be assigned to the operation. The resource type must be known in the system. Predefined resource types must be chosen from the selection menu. Additional resource types can be defined when customizing HYDRA. For documents, the resource type to be entered here must be DOC.

Resource

Enter the resource number (material number) of the production resource or tool.

Designation

Here, you can enter a name for the production resource.

Comment 1/ Comment 2

These are comment fields.

Required quantity/ unit

Resource quantity required to carry out the operation. When planning the operation in HYDRA shop floor scheduling (HLS), this number of resources is entered in terms of capacities. The quantity unit is only used as a comment.

Please note: In HYDRA shop floor scheduling (HLS), the quantity 0 is interpreted implicitly as quantity 1.

Path

When identifying a document as a production resource, the logical reference to the path is to be defined in the path configuration (menu: File > System administration > Paths). No path must be stored for DNC resources; it is determined based on the path stored for the resource type. The field should be left empty for all other production resources.

File

When identifying a document as a production resource, the file name (including file extension) is to be entered here.

No file name must be stored for DNC resources; it is determined based on the file name defined for the resource. The field should be left empty for all other production resources.

Modified by/ date/ time

Editor as well as the date and time the last change was made.



Please note with regard to documents: If a new document is assigned to an operation a file is only uploaded automatically, in case a file has been selected using the file selection dialog. The file selection dialog can be opened by the button next to the "file name" field.

In this case, the path of the file that is loaded onto the server is displayed below the input field for the file name. The upload is performed automatically while saving.

No file can be uploaded if the file name is entered manually.

The corresponding data record is created anyway even if an error occurs during the upload.

Toolbar



Edit operations

Calls up the application [Edit operations](#).



Edit orders

Calls up the application [Edit orders](#).



Order information

Calls up the application [Order information](#).

12 Edit Order Network

Summary

Menu	Order Management → Order Management → Edit Order Network
Transaction code	ednet
Function authorization	ednet

Usage

With the support of this application, you can create dependencies for orders beyond the existing operation sequence. These dependencies are referred to as relationships.

Keep in mind that only the end-start relationships can be created. These are relevant for both planning and for data entry. Enter the MES order number (combined order/ OP number) during data entry.

Requirement

The linked orders, including all of their operations, must exist in the system.

Selection criteria

The application provides the following selection criteria:

Order

The relationships are displayed for the selected order number.

OP

The relationships are displayed for the selected operation.

Predecessor/ successor/ predecessor and successor

Only the relationships relating to the selection are displayed.

Toolbar

This application only allows relationships to be created or deleted.



Any relationships created by the system automatically (origin = "S") may not be deleted by the user.

Field descriptions

Predecessor

Order number of the preceding operation

Preceding OP

Operation number of the preceding operation

Successor

Order number of the succeeding operation

Succeeding OP

Operation number of the succeeding operation

Relationship

Only the end-start relationships ("ES") can be created in the setup process.

Origin

Relationships created manually or explicitly via the interface are created using "E" = externally created.

The relationships created by the system are marked with "S".

Active

In principle, relationships are always active. Relationships created due to alternative sequences are the exception. Relationships of inactive alternative sequences are marked as inactive.

Relevance

The system differentiates between relationships for planning and relationships for data entry.

P Relationship is only relevant for planning.

V Relationship is only relevant for data entry.

X Relationship is neither relevant for planning nor for data entry.

<empty> Relationship is relevant for planning and for data entry.

Explicitly set relationships can only be created with relevance =<empty>.

13 Work Plan - Edit Orders

1.1 Summary

Menu	Order management → Routing management → Work plan - Edit orders
Transaction code	edwor
Function authorization	edwor

The "work plan - edit orders" application provides the user with a comfortable option to create and change work plans as well as to generate orders from work plans. Consequently, real orders may be generated from the work plans that are also often referred to as "empty envelope" for orders.

Selection criteria

The application provides the following selection criteria:

Work plan

A specific work plan may directly be selected by entering the work plan number.

Order type

The combo box allows for work plans for specific order types to be selected. Multiple selections are possible.

Article

The article field allows for work plans for a specific article to be searched. Wildcards may be used.

Sales order, project number, planned order

These fields are stock fields of the order header that are available for selections. Wildcards may be used.

Customer designation

The "customer designation" field may be used for searching, provided that work plans are created for individual customers. Wildcards may be used.

Field descriptions

Order header fields are described [here](#)



Please note

- The table only provides selected data:
 - Work plan

- Order type
- Article
- Article designation
- Target quantity (B)
- Target scrap (B)
- Unit (B)
- Customer designation
- Sales order
- Planned order
- Project number
- The below-mentioned values cannot be kept in the work plan order
 - Start of basic date
 - End of basic date
 - Scheduled start
 - Scheduled end

Editing functions

Please use the available buttons to create or edit work plan orders.

Provided that a responsibility area has been defined for the order, maintenance (editing) of data depends on how the view, insert, change, delete options are set in the configuration of responsibility areas or responsibility profiles.

Toolbar



Generate order

Function authorization: or.generate

An order may be generated from the currently selected work plan by calling this function. For further information on this, please refer to the section entitled [generate order](#).



Edit order long texts

Function authorization: edwortx

Starts the application [Work plan - edit order long texts](#).



Edit order sequences

Function authorization: edwseq

Opens the application [Work plan - edit order sequences](#).



Edit operations

Function authorization: edwop

Opens the application [Work plan - edit operations](#).

Generate order

Please proceed as follows to generate an order from the work plan:

- Choose the work plan, from which you would like to generate an order, from the table.
- Open the function using the button . The "generate order" dialog opens.
- In the "order" field enter the order number that is to be assigned to the order to be generated. The field can be left empty if numbers are assigned automatically for the order type (customizing).
- In the dialog the input fields are pre-assigned to work plan data. If required, change or amend the field values.
- Confirm the dialog by clicking .

An order is now generated from the work plan. Then the application "[edit orders](#)" including the new order opens by default.

Please note that the "edit orders" application is opened in a new window every time a new order is generated. For this reason, we recommend to close the application before generating a new order.

The below-mentioned configuration in INI configuration/INI data configuration can suppress opening of the application "edit orders". In this case and once the order has been generated successfully, a popup dialog opens stating that "order xxx" has been created". This message can be confirmed by clicking "OK".

Menu: System administration → System settings → INI configuration/INI data configuration

Name: BDE

MOC user: 0

Comment: Settings for Shop Floor Data Collection

Section: EDWOR

Key: GENERATE_ORDER

Value: SUPPRESS_EDOR

Active: 

Comment: Suppress automatic call of application "edit orders"



Please note

- The order cannot be generated if a responsibility area is indicated for which the user is not authorized.
- The article number is taken over to the operations of the order.
- On the one hand the order quantity as well as the unit are taken over to the order header as (basic) target quantity or target unit, and on the other hand, to all operations as basic target quantity or target unit.
- The entered quantity is taken over 1:1 as primary quantity, if the primary quantity unit (primary unit of entry) of the operations matches the unit that is specified as quantity unit above. If conversion factors are defined for the operation of the order or work plan to be copied, they will also be used for the calculation of the primary quantity. In case, no conversion factors are defined and the base quantity unit and primary quantity unit are different, an attempt is made to convert the base quantity into the primary quantity unit using an internal conversion table (HYDRA customizing). In general, this procedure also applies for the secondary quantity and tertiary quantity.

14 Work Plan - Edit Order Long Texts

1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit long texts of orders
Transaction code	edwortx
Function authorization	edwortx

The "work plan - edit long texts of orders" function allows for additional texts to be collected for the order. There is only one long text at most for each order.



Please note

- Long texts can also be transferred using the HYDRA info interface (record type "AI"). For further information on the interface, please refer to the corresponding interface documentation.
- The terminal shows the long texts relating to operations only.

Selection criteria

The application provides the following selection criteria:

Work plan

The long text of a specific order of the work plan can be selected by entering the work plan number.

Field Descriptions

The fields of a long text pertaining to orders are described [here](#).

Editing functions

Please use the available buttons to create or edit long texts of work plan orders. A copy function for order long texts is not planned.

Toolbar



Edit orders

Function authorization: edwor

Opens the [Work plan - edit orders](#) application for the currently selected data record.

15 Work Plan - Edit Order Sequences

Summary

Menu	Order management --> Routing management --> Work plan - Edit order sequences
Transaction code	edwseq
Function authorization	edwseq

Operations are grouped in sequences to summarize them within an order. Production uses this information as an orientation tool to process each operation. Within the sequence, the operations are processed in sequence one at a time. By linking several sequences within the order, network-type structures can be illustrated

For further information on this, please refer to the document entitled ["edit order sequences"](#).

Selection criteria

The application provides the following selection criteria:

Work plan

The order sequences of a specific work plan may be selected by entering a work plan number.

Field descriptions

The fields of a sequence are described [here](#).

Editing functions

Please use the available buttons to create or edit work plan sequences. A copy function for order sequences is not planned.



If the "sequence" field is not shown in the editing dialog, the sequence number length is 0 in the basic parameter settings. Please contact MPDV.

Toolbar



Edit orders

Function authorization: edwor

Opens the application [work plan – edit orders](#) for the currently selected data record.

16 Work Plan - Edit Operations



1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit operations
Transaction code	edwop
Function authorization	edwop

The "work plan - edit operations" application provides the user with a comfortable option to create or change operations for work plans.

Consequently, real orders may be generated from the work plans that are also often referred to as "empty envelope" for order orders.

Selection criteria

The application provides the following selection criteria:

Work plan

Operations of a specific work plan may be selected by entering the work plan number.

OP

The view may be limited to an individual operation of a work plan.

Sequence

This field is only relevant if sequences are used in HYDRA. In this case, the selection may be restricted to a specific sequence.

Field Descriptions

Operation fields are described [here](#)

Editing functions

Please use the available buttons to create or edit work plan operations.

The [order \(header\) of the work plan](#) must already exist in order to be able to create an operation.



Operations may only be copied within the work plan.



This table is to be used to document warnings/alerts (template).

Toolbar



Edit orders

Function authorization: edwor

Opens the application [Work plan - edit orders](#) for the currently selected data record.



Edit order sequences

Function authorization: edwseq

Opens the application [Work plan - edit order sequences](#).



Edit operation long texts

Function authorization: edwoptx

Opens the application [Work plan - edit operation long texts](#).



Edit components

Function authorization: edwcomp

Opens the application [Work plan - edit components](#).



Edit production resources and tools

Function authorization: edwres

Opens the application [Work plan - edit production resources and tools](#).

17 Work Plan - Edit Operation Long Texts



1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit long texts of OP
Transaction code	edwoptx
Function authorization	edwoptx

The "work plan - edit long texts of operation" function allows for additional texts to be collected for the operation. There is only one long text at most for each operation.



Please note

- Long texts can also be transferred using the HYDRA info interface (record type "AI"). For further information on the interface, please refer to the corresponding interface documentation.
- The terminal shows the long texts relating to operations.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The long text of a specific operation of the work plan can be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Field Descriptions

The fields of a long text pertaining to orders are described [here](#).

Editing functions

Please use the available buttons to create or edit operation long texts of the work plan. A copy function for operation long texts is not planned.

Toolbar



Edit operations

Function authorization: edwop

Opens the application [Work plan - edit operations](#) for the currently selected data record.

18 Work Plan - Edit Components



1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit components
Transaction code	edwcomp
Function authorization	edwcomp

The "work plan - edit components" application allows for the material components, which are required to produce the article in the current manufacturing level (current operation), to be displayed and edited.

Normally, these components are transferred to HYDRA using an interface from the higher-tier ERP system, as these components are already defined in the ERP work plan.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The components assigned to a work plan operation may be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Enter the whole MES work plan number if you would like to view the components assigned to a specific operation.

If you would like to view the components of all operations of a work plan only enter the work plan number, followed by "*".

Field Descriptions

The fields of a production tool and resource are described [here](#).

Editing functions

Please use the available buttons to create or edit work plan components. A copy function for components is not planned.



Please note that the BOM item must be **unique** within the operation if HYDRA-MPL is in use!

Toolbar



Edit operations

Function authorization: edwop

Opens the application [Work plan - edit operations](#).



Edit orders

Function authorization: edwor

Opens the application [Work plan - edit orders](#).

19 Work Plan - Edit Production Resources & Tools



1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit production resources & tools
Transaction code	edwres
Function authorization	edwres

The "production resources & tools" application allows for the resources, which are required to produce the article in the current manufacturing level (current operation), to be displayed and edited.

Production resources and tools may be, for example, tools, documents, NC programs, etc.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The production resources and tools assigned to a work plan operation may be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Enter the whole MES work plan number if you would like to view the production resources & tools assigned to a specific operation.

If you would like to view the production resources & tools of all operations of a work plan only enter the work plan number, followed by "***".

Field Descriptions

The fields of a production tool and resource are described [here](#).

Editing functions

Please use the available buttons to create or edit production resources & tools of work plans. A copy function for production resources & tools is not planned.



If the tool and resource management module (HYDRA-WRM) is in use, the first production resource and tool that is *not* of the resource type "DNC" or "MAT" is taken over into the "tool" field of the [operation](#). In addition, the "tool" field is checked whether it already includes a value, when inserting a production resource and tool that is not of the "DNC" or "MAT" resource type. If this is not the case, this component is taken over. For this reason, it is recommended to insert the "main production resource & tool" at first in the list of production resources and tools.



Please note with regard to documents: If a new document is assigned to an operation a file is only uploaded automatically, in case a file has been selected using the file selection dialog. The file selection dialog can be opened by the button next to the "file name" field.

In this case, the path of the file that is loaded onto the server is displayed below the input field for the file name. The upload is performed automatically while saving.

No file can be uploaded if the file name is entered manually.

The corresponding data record is created anyway even if an error occurs during the upload.

Toolbar



Edit operations

Function authorization: edwop

Opens the application [work plan - edit operations](#).



Edit orders

Function authorization: edwor

Opens the application [work plan - edit orders](#).

- →→→→

20 Data Structure of Orders

Each of the fields for an order header are described below. The actual sequence of the editing dialogs and analyses/ overview may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

General index tab

Order and/or work plan

The order number or rather the work plan number is an upper-level number, under which each of the operations is compiled.

Order type

Order types are issued to structure the orders in accordance with their use. Each order type includes various control information that is decisive when managing orders.

The standardized order types available in the system are described in the glossary. Additional order types can be defined when customizing the system.

Article

Material numbers/ reference numbers/ article numbers of the (final) article to be produced with this order. If no article is entered for the operation, it is transferred to the operations.

Article designation

Name of the article. Any changes to the article designation are assumed redundantly in all operations of the order. They cannot be entered by operation.

Customer designation

Customer name.

Sales order

Sales order number

Sales order item The line item number can be displayed after the sales order number on the sales order that this order refers to.

Priority

The priority function can be used as a control tool for the order. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

This is the value that is used to control a color code designation in the graphic planning board (shop floor scheduling) via the operation bar and which for planners is a graphic means to show the priority of the operations. The displayed colors are assigned to the priorities in the shop floor scheduling in the settings for the graphic planning board.

During the customizing process, it can be determined based on the order type

- whether the priority of the order header is transferred unchanged to each of the operations
- whether priority management should be turned on.

Order index

The order index can be seen as an alternative to the priority. It can, for example, (if configured accordingly) be included when sorting the graphic detailed planning.

The order index tab is numerical with a valid value range (-999.9 to +999.9).

Target quantity

Quantity specification for the production order in base quantity unit. The indicated target quantity may include a target scrap quantity that might have been entered.

Target scrap

Planned scrap for the production order in base quantity unit. The indicated scrap quantity can be considered as part of the transferred target quantity.

Unit

Quantity unit of the order for the (final) article to be produced. It establishes the degree of comparability with scrap from different operations, for example, and is therefore redundantly included as a base quantity unit in each operation.

Material type

Material type of the (final) article to be produced. If no material type is entered, MES inserts the value "SYSTEM" here.

Batch number

The batch number reserved for the order; is generally provided by the ERP system.

Dates index tab

Basic date start

The basic start date for the order. It is generally specified by the ERP system.

Basic date end

The basic end date of the order. It is generally specified by the ERP system and, in some cases accounting for buffers, it is based on the date required/ delivery date set in the ERP system.

Scheduled start time

Scheduled start date; result of the lead time scheduling as compared to infinite capacities.

If the scheduling is run outside of the system, the scheduled lead times in the order header should be applied. If the scheduling is run in MES, these fields are overwritten.

Scheduled end time

Scheduled end date; result of the lead time scheduling as compared to infinite capacities.

If the scheduling is run outside of MES, the scheduled lead times in the order header should be applied. If the scheduling is run in MES, these fields are overwritten.

Scheduling type

The scheduling type describes whether the order is scheduled forward (V) or backward (R) during lead time scheduling in MES. If scheduled forward, the order is scheduled on the basis of the basic start date specified in the ERP system. If it is scheduled backward, it is scheduled on the basis of the basic end date specified in the system.

If no scheduling type is set in the order, the scheduling type defined in *Basic Settings* is used.

Reduction strategy

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available, then MES will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the wait times and the transport time.

The steps to take to configure the reduction strategies are described in document hls-bk.doc in the chapter Reduction Strategies and are performed as part of the customizing process.

Assignment index tab

Order group

Each order can be assigned to one order group. Based on this order group, the priorities are managed, which means the proportion of valid priorities is each assigned based on this kind of order group. This makes it necessary to consider the order group in the order as a mandatory field and to check it against a preset value range. Otherwise, it could not be assured that the priority assignment can be checked.

Note for SAP users

The term order group corresponds to the SAP production scheduler. For a consistent data exchange between SAP and HYDRA, the possible production schedulers in SAP should be entered at the same time that order groups are entered in MES.

MRP controller

The MRP controller for the order, which, for informational purposes, can be assumed in MES and which is displayed in the console. No value pool is managed in MES in this regard.

Project number:

Project order number

Planned order

Planned order number, e.g. in serial production

Cost object

Cost object number

Work plan

Work plan number of the work plan that served as the template for generating the production order.

Work plan version

Version number of the work plan that served as the template for generating the production order.

BOM version

Version of the bill of materials assigned to the production order.

Production version

Production version on which the order is based. It is currently only populated when planned orders are assumed via the HKMPP-REM interface.

Control cycle

ID of the control cycle / supply relationship for which a kanban order has been generated.

Inspection order

Inspection order/ inspection batch number for the order

Sample type

Type of sample for the order

Calculation index tab

The calculation index tab includes additional data fields in which calculation-related values or information can be stored. These entries are for information purposes only.

Machine costs

Calculated value for the machine costs that are incurred in the production of this order.

Labor costs

Calculated value for the labor costs that are incurred in the production of this order.

Material costs

Calculated value for the material costs that are incurred in the production of this order.

Other costs

Calculated value for other costs that are incurred in the production of this order.

Material value

Calculated value of the produced final article for each base quantity unit.

Scrap value

Calculated scrap value for each base quantity unit.

User fields index tab

User fields offer the opportunity to enter additional customer-specific information in MES in addition to the fields available in MES Standard. The order-related user fields are displayed in the order information. The user field index tab is shown here for the order header, which, in addition to the user field key, also displays the defined user fields and includes a designation and unit of measure. There are eight sub-index tabs in the index tab, which each have eight additional user fields. The so-called user field key determines which user fields they are and what their purpose is.

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next.

User fields

The following maximum user fields can be available in the order header (AUNR object type) if customized accordingly:

Field ID/Index	Field data type	Number of fields
1 - 6	Date	6
7 - 22	Numeric, time, duration	16
23 - 28	Decimal value	6
29 - 44	Text field, length 1	16
45 - 50	Text field, length 10	6
51 - 64	Text field, length 20	14
65 - 66	Text field, length 40	2

User field keys are defined in coordination with the customer during the customizing process.

Administration index tab

The administration index tab includes technical information on the data record. This index tab is not available in the dialogs "Insert" and "Copy".

Created by

User who entered the order.

Created on:

Time and date the order was entered.

Editor

User who made the last modification in the order header.

Modification

Time and date when this modification was made.

Transfer from

Here, you can enter the source from where the order was transferred.

Transfer time

If the order was transferred from the ERP system (PPS=J), the transfer time and date is automatically set to the time and date on which it was stored in MES.

Modified HYDRA

Information that states that the order was modified in MES. This identifier is automatically set at "J", if a modification is made in MES.

Modified PPS

Information that states that the order was modified in the ERP system. This identifier is automatically set at "J", if a modification is made via an ERP interface (PPS=J). The identifier is not reset.

Deletion flag

Used for internal processing purposes. Cannot be modified.

Responsibility area

If a responsibility area is entered here, the user must have been authorized to display and edit orders and/or work plan orders.

21 Order Long Text Data Structure

Each of the fields for an order long text are described below. The actual sequence of the editing dialogs may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

Order/ work plan

Order or work plan for which a long text is defined.

Short text

Short version of the long text, which is shown in the application list.

Long text

Actual long text, which is not shown in the application list.

The text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

22 Data Structure of Order Sequences

Each of the fields for a sequence is described below. The actual sequence of the editing dialogs may deviate from the one illustrated here. Information about sequences can be found in the document [edit sequences](#).

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

Order/ work plan

Order for which the sequence is defined. A sequence can only be set up for an order, if the order header already exists in MES.

Sequence

Identification of the sequence within an order.

Please note: The standard sequence always has the sequence number 0.



If the "sequence" field is not shown, the sequence number length is 0 in the basic parameter settings. Please contact MPDV.

Designation

Description of the sequence.

Sequence category

S = Standard sequence

There is only one standard sequence for every order; it cannot be deleted.

P = Parallel sequence to the standard sequence

There can be several parallel sequences for each order.

A = Alternative sequence to the standard sequence

There can be several alternative sequences for each order.

Please note:

The sequence category cannot be edited after a sequence has been set up!

Active

The qualification "Active" is only relevant for alternative sequences:

J = Active

N = Not active

If a new alternative sequence is set up, it is set as **not active**.

For standard sequences and alternative sequences, this qualification is always set to **active**.

Orientation

If there are several parallel sequences, the lead times generally vary in length. This creates time buffers in the sequences. The orientation function controls whether these buffers are at the beginning or the end of the sequences. The following options are available:

F = Earliest due date

If the sequence is set for the earliest date, the buffer will be at the end of the sequence.

S = Latest due date

If the sequence is set for the latest date, the buffer will be at the beginning of the sequence.

N = Not relevant; this is the case for standard sequences and alternative sequences.

If there are several parallel sequences for a given standard sequence, the orientation of the standard sequence is used for all segments of the standard sequence for which parallel sequences exist.

Version

Change number/version; for information purposes only.

Branch operation

Operation number of a standard sequence operation,

before which a parallel sequence should branch off, or

from which on an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch operation of the standard sequence must already exist in the orders on hand. When a sequence is handed over via an interface, a valid order number also must be handed off (there is no plausibility check).

Return operation

Operation number of a standard sequence operation,

after which a parallel sequence should branch off, or

up to which an alternative sequence should be replaced.

This is a mandatory field for parallel and alternative sequences. For a standard sequence, this field must remain empty.

If manually setting up an alternative or parallel sequence, the branch-off operation of the standard sequence must already exist in the orders on hand. When a sequence is handed over via an interface, a valid operation number also must be handed off (there is no plausibility check).

Reference sequence

The reference sequence determines the sequence in the order that the reference operations (branch and return) refer to. This is **always** the standard sequence (sequence number 0).

This is a mandatory field for parallel and alternative sequences. The standard sequence must already exist.

For a standard sequence, this field must remain empty.

23 Data Structure of Operations

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Each of the fields for an operation is described below. The fields are structured in this case using the index tabs. The actual sequence may deviate from the one illustrated here.

In order to simplify matters, the term order will generally be used, regardless of whether an order or a work plan is being discussed. Only in examples in which it would make sense for the overall understanding to differentiate between the two will we use the term work plan.

General tab

Order / work plan

The order number or rather the work plan number is an upper-level number, under which each of the operations is compiled.

Sequence

The sequence number is the number of operation sequences in use.

OP

The operation number is the number listed below the order used to identify the operation.

Split

The split number

OP designation

Name of the operation; generally simply a short description of the activities that will be performed.

Article

Reference/item number of the article or material that is produced with the operation. If no article is entered, the article from the field with the same name in the order header is assumed here.

Material type

Material type of the article that is to be produced in this particular production step. If no material type is entered, the value from the field with the same name in the order header is assumed here.

Priority

The priority can be used as a control tool. The priority is a single digit, numeric value. The value increases in ascending order ("0" = lowest priority, "9" = highest priority).

Depending on the [order type](#), the priority may be configured so that they either reference the operation or the order. Choosing the latter will mean that the priority of the operation is assumed from the order header.

Planned on

This identifier describes whether the operation is located in the group pool or if it is planned on a specific workplace / machine. The operations are planned in MES either within the graphic detailed planning or in the order sequencing stage.



Entering or deleting the workplace at a later time will NOT automatically change this identifier.

Planned workplace

If the identifier **planned for workplace** is set, this means that the operation is planned for the workplace entered here. If the input field is empty, the operation is not planned for any workplace.

Please note: When an operation is logged in, this field is automatically set to the workplace at which the operation was logged in. Doing so will overwrite any (in some cases a different) workplace for which the operation was planned up until then. As a result, the OP is implicitly re-planned.

Group

(Planned) station / machine group designated for producing the operation. It is meant as a planning criterion for group-oriented planning and in the graphic detailed planning.

When an operation is logged into a (different) workplace, its group will be updated, if necessary.

If, due to logging in the operation, there is a change to the group for which the operation was planned up until now, NONE of the values are assumed from the template (this only happens if modifications are made manually in the editing function).

Fixed

This identifier specifies whether an operation is set as fixed during the planning process.

Before running automatic planning, the capacities (workplaces) are completely released with the exception of the fixed operations. Fixed operations that are still set in the past are moved to the right and set to "now" at the earliest plus a planning lead time. Any operations planned for the future remain dispatched without changes.

Material

The first resource of the type "Material" (ID: "MAT") available in the component list is assumed in this field. This is the "most important" input material.



This field is used in planning, for example, when applying the [setup change list](#) or in graphic detailed planning when planning equipment setup changes. It is of no significance to processing as part of the material and production logistics (MPL):

Color

You can enter the color of the main input material or the article planned for production here. This field is used in planning for example in the [setup change list](#) or when planning equipment setup changes.

Tool

When using the tools and resource management module, the first resource available in the production resource / component list that is **not** designated with the resource type "DNC" or "MAT" is assumed here.

To this end, when a component is being entered that does **not** have a resource type "DNC" or "MAT", the system checks whether this field already includes a value. If no value is entered, this component is assumed. For this reason, we recommend to first input the "main production resource" in the production resource list.



This field is considered in the graphic detailed planning to determine production variants. However, the production resources stored in the operation are relevant when checking capacities.

By default, the field is of no relevance for processing in tools and resources management.

DNC

When applying the production facility management, the first resource available in the production resource / component list that has a resource type "DNC" (ID: "DNC") is assumed. To this end, when a component with a resource type "DNC" is entered, the system checks whether there is already a value in this field. If no value is entered, this component is assumed.



In the system, this field is mainly used as a comment. By default, it has no significance for DNC processing.

Upload number

The purpose of the confirmation/upload number is to identify an operation. This is a numeric value that is used for postings as an alternative to the combined order/OP number.

Examples of applications

- Most of the time, a bar code is hard to read if the order / OP number is long (for example when using bar code guns with a limited scanning range);
- if the space available on the work document is not large enough.

Please note: The length of the input field depends on the settings made for "Length of upload number" in the basic parameter settings. If no length is specified there, the field is shown across the whole width of the application.



This field must be left blank for work plan operations.

Authorization

An authorization identifier that indicates whether a user is authorized to log this operation on / off. This involves cross-checking the identifier *OP postings* in the [HR master data](#).

Cost type

The cost type to be posted when executing this operation, for example in an overhead cost operation / order. At the moment, this field is only used as a comment.

Cost center

The cost center to be debited when executing this operation, for example in an overhead cost operation / order. At the moment, this field is only used as a comment.

Dates tab

The following dates are results that were calculated and executed during lead time scheduling. Lead time scheduling is triggered by certain events and runs asynchronously in the MES system.

Scheduled start time

Scheduled start time of the operation as a result of the lead time scheduling compared to infinite capacities.

As a rule, a fixed operation is never changed. If, based on the scheduling situation, a date cannot be maintained, the operation is rescheduled, but it remains fixed.

Scheduled end time

Scheduled end time of the operation as a result of the lead time scheduling compared to infinite capacities.

Earliest start

Earliest start date (EST) of an operation as a result of forward scheduling during lead time scheduling as compared to infinite capacities or specified by PPS.

Earliest end

Earliest end date (EET) of an operation as a result of forward scheduling during lead time scheduling as compared to infinite capacities or specified by PPS.

Latest start

Latest start date (LST) of the operation as a result of backward scheduling during lead time scheduling as compared to infinite capacities.

Latest end

Latest end date (LET) of the operation as a result of backward scheduling during lead time scheduling as compared to infinite capacities.

Buffer time

The buffer time is determined based on the difference between the latest start date (LST) and the earliest start date (EST) for an operation

The sum total of the buffer times of all operations is stored in the order (header) in the field *OP buffer*.

Reducible time

If it turns out during scheduling that the lead time for a given order is longer than the allotted time available (basic end date exceeded), then MES will attempt to take reduction measures to shorten the lead time accordingly. Reducible times are the wait times and the transport time.

This value indicates how many (more) hours can be reduced from the lead time of an order. The time for an operation that can still be reduced is equal to the difference from the current time and the minimum time, each for the wait time and the transport time. These differences are displayed here as totals.

Information on how reduction strategies are configured is provided in the document entitled [reduction strategies](#).

Planned start

Planned start date for the operation.



Logging in an operation that has not yet been planned will not result in the time for the first operation start to be interpreted as a planned start and therefore storing it as such here.

Planned end

Planned end date for the operation.

The planned dates (planned start/planned end) are set:

- by HYDRA Shop Floor Scheduling (HLS) upon saving
- by the Graphic Order Sequencing (GAV) upon saving
- by manual data maintenance (client)
- by the interface

The following logic applies for inserting and/or editing an operation (manual data maintenance or by interface):

- Insert/copy operation
 - If the planned start is empty, it will be assigned to the earliest start date by default.
 - If the planned end is empty, it will be assigned to the latest end by default.
 - In both cases, however, the operation is not planned (automatically), i.e. the operation can still be removed from planning.
- Change operation
 - In this case, processing depends on the "planning function" flag of the workplace:
 - N (no planning):
 - If the planned start is empty, it will be assigned to the earliest start date by default.
 - If the planned end is empty, it will be assigned to the latest end date by default.

- In any other cases, the planned dates will not be set automatically through processing.

Quantities tab

Generally, quantities can be listed in four different units for an operation. Provided for each quantity unit is the target quantity and quantity unit. There is also the option to provide a calculated "estimated scrap". These quantities can be specified by the PPS system or, if the target quantity update is activated, may be the result of the produced quantity from the previous operation.

The letters in parentheses behind the field descriptions provide information about the particular quantity type.

- (P) Primary quantity unit
- (S) Secondary quantity unit
- (T) Tertiary quantity unit
- (B) Base quantity unit

Target quantity (P) / unit / target scrap quantity (P)

The primary quantity is a unit in which entries are made at the terminal (primary input quantity).

The indicated target quantity may include a target scrap quantity that might have been entered.

Send-ahead quantity

In order to illustrate any overlapping, a minimum send ahead quantity (in the primary quantity unit) can be defined for an operation. The following operation can be started as overlapping if at least the send ahead quantity is reported as completed. In addition to the validation check relating to the send ahead quantity that is performed when data is entered (operation log in), any overlapping is also accounted for during scheduling and during detailed planning.



The relevant configuration needs to be enabled at the order type to check the minimum send ahead quantity, when logging OPs on. The system supports overlapping operations with respect to the minimum send ahead quantity (or the lead time). This is enabled at the processing code while customizing the system.

When checking the minimum send ahead quantity, the system only accounts for the recorded yield that was entered up until now (primary quantity unit).

Quantities are not converted. For this reason, please pay attention to the fact that adjacent operations have the same primary quantity unit.

Example:

Operation 0100	Target quantity 1000	Send ahead quantity 50
Operation 0200	Target quantity 1000	

If checking is enabled, operation 0200 can only be logged on as soon as a yield (in primary quantity unit) of at least 50 has been uploaded/confirmed for operation 0100.



Checking does not include the operation status of the preceding operation. The current operation cannot be logged on, in case the preceding operation has already been finished, but the send-ahead quantity has not yet been reached.

Target quantity (S and T) / unit / target scrap quantity (S and T)

The secondary and tertiary quantity are considered optional, variable units (for example within the reel-based MES solution - RF).

The indicated target quantity may include a target scrap quantity that might have been entered.

Target quantity (B) / unit / target scrap quantity (B)

The base quantity unit is an objective description of the material used in an order, which means it represents a level of comparison of, for example, scrap from different operations. The base quantity unit is in effect the quantity unit shown in the work order header. Generally, conversions (for example when target quantities are updated) are made using the base quantity unit.

The indicated target quantity may include a target scrap quantity that might have been entered.



Please give special attention when using a quantity type that the relevant quantity unit is set correctly.

The quantities are only converted based on the conversion factors in the index tab "quantities" if the relevant values that need to be recalculated have been set to "empty" (not "0"). Quantity fields that contain values will not be recalculated.

Conversion factors

Conversion factors are used to convert primary, secondary and tertiary quantities to a base quantity. These are used, for example, at the time the target quantities are updated.

In order for the conversion factor to also be able to use a decimal value (meaning a figure with decimal places), they are each illustrated using their relevant numerator and denominator.

Example

- Base quantity unit: Square meter M2
- Primary quantity unit: Piece PCE
- 1 piece = 2 square meters.

In this case

- the numerator is defined as the primary quantity 2 and
- the denominator is defined as the primary quantity 1

If no (valid) conversion factor exists, the system will attempt to convert using conversion formulas (this requires that formulas were defined accordingly during the customizing process).

Overdelivery/ Underdelivery

All quantity postings made at the time of partial confirmation/upload, interruptions or when logging off the separate operations are subject to an operation-related verification to check for overdelivery. When an operation is logged off, an additional verification is run to check for underdelivery.



Further information on overdelivery/underdelivery checking can be found in the document entitled [MBL_PC_UnderOverDeliveryOverview.pdf](#).

Underdelivery (%)

Value shown as a percentage by which the quantity reported back may deviate from the target quantity (primary quantity unit). The value is only assumed from the processing code if the value was not explicitly transmitted via the ERP interface.

Example:

Target quantity of the operation: 120 items

Underdelivery: 84%

The actual quantity must not fall below 70 items.

Overdelivery (%)

Value shown as a percentage by which the quantity reported back may deviate from the target quantity (primary quantity unit). The value is only assumed from the processing code if the value was not explicitly transmitted via the ERP interface.

Example:

Target quantity of the operation: 120 items

Overdelivery: 168%

The actual quantity must not exceed 140 items.

Overdelivery reaction/ underdelivery reaction

If the limits in the fields overdelivery or underdelivery are exceeded, a warning or an error message may be issued in response. Possible values:

"empty" No reaction

W Warning

X Error.

When setting a warning as a reaction, you can override the validation check by entering a deviation reason.

If Error is set as a response, you will not be able to override the validation check.

Please note: The messages can only be processed further at Windows terminals. When using DOS terminals, the response "W" is interpreted as an error.

Unit quantity

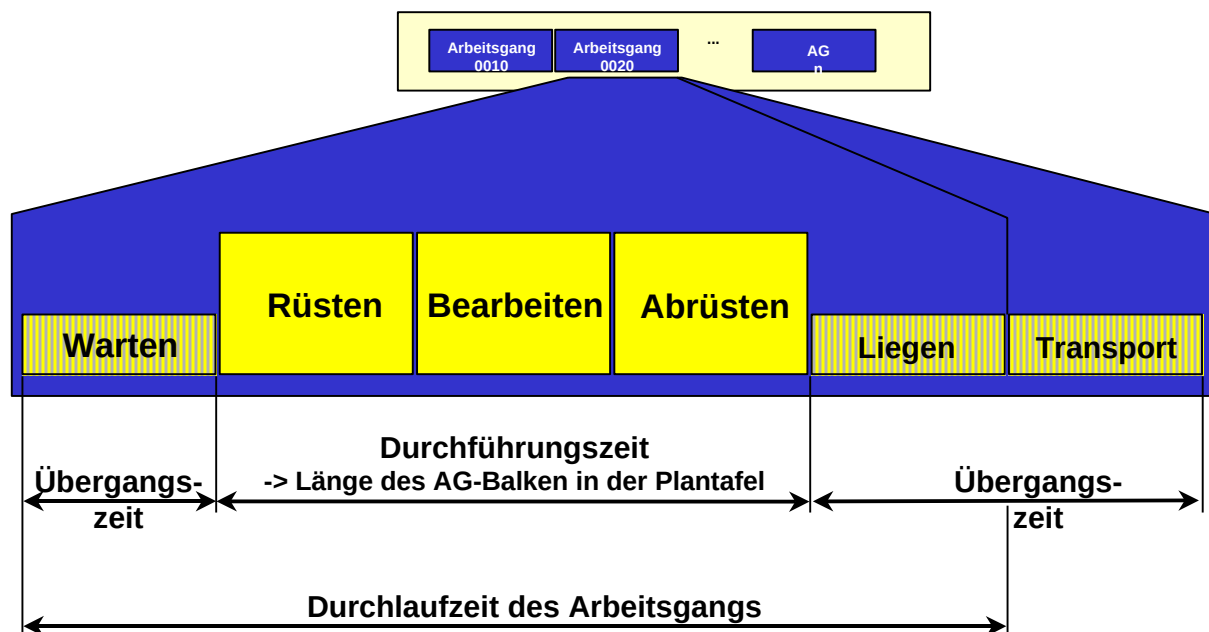
Quantity that is referenced in the operation specifications. In MES (customized), here the ERP base quantity can be assumed and be referenced in formulas used to calculate processing times.

The unit of the unit quantity must be a primary quantity unit. The system does not perform an automatic conversion if the quantity units do not match.

As opposed to the basic quantity in ERP, there is no other meaning or use in MES.

Durations / target times tabs

The illustration shown below provides an overview of the chronological structure of an operation in MES (in house production).



Target setup time

The target setup time is the time required to prepare a workplace for the operation, for example the time needed to mount the necessary tools or to set the machine using the relevant definitions ("setup time"). During this time, the workplace's capacity is shown as in use

The target setup time is transferred from the PPS system or custom formulas calculated from specified values can be defined; in this case, the formula must be entered in the field "Setup time formula".

Additional setup time

The additional setup time is set at the operation using graphic detailed planning if a setup change matrix was entered and if any such results from planning.

The additional setup time can also show a negative value.

Target processing time

The processing time is the time needed to process the material during an operation. During this time, the workplace's capacity is shown as in use. The processing time depends on the order quantity; it does neither include the setup time nor the dismantling time.



The processing time is not used in the graphic detailed planning, but instead the processing or rather the remaining run time is calculated dynamically using the formula entered in the field "Formula RRT 1".

The target processing time is transferred from the PPS system or it can be calculated using a custom formula based on predefined values. In this case, the formula must be entered in the field "processing time formula". What needs to be remembered here is that the processing time is calculated using the same principle as the one used to calculate the remaining run time (field "Formula RRT1").

Planned dismantling time

The planned dismantling time (teardown/retooling time) is the time needed to reset the workplace back to its original state after the operation has been completed. This may require some tasks such as dismantling tools or performing some cleaning work. During this time, the workplace's capacity is shown as in use.

The planned dismantling time is transferred from the PPS system or it can be calculated using a custom formula based on predefined values. In this case, the formula must be entered in the field "Dismantling time formula".

Planned delivery time

There is only one time component for external operations, the delivery time. It is matched to the Gregorian calendar. The performance level has no relevance.

External processing

If this identifier is set, the operation is one that is performed externally. External operations are generally not planned, but only scheduled. In terms of scheduling, the lead time results only from the delivery time for these kinds of operations.

If this identifier has not been set, it will be considered an in house operation. The capacity requirements are calculated for these kinds of operations based on processing times:

- Target setup time
- Target processing time
- Target dismantling time.

On the other hand, the following processing times are used for scheduling (lead time scheduling):

- Planned waiting time
- Target setup time
- Target processing time
- Target dismantling time
- Planned idle time
- Planned transport time.

Formula RRT1 / Formula RRT2

Formula RRT1 is used to describe the calculation of the remaining run time (RRT) for an operation. It is used in the graphic detailed planning.

Unless otherwise specified or defined, the formula to be entered here is the RRT value. In this case, the remaining run time is calculated as follows:

$$(\text{Target cycle} / 1,000) * (\text{primary target quantity} - \text{the yield recorded up until now}) / \text{partitioning}$$

The RRT 2 formula is an optional one that is used to calculate any remaining run time that might deviate from RRT 1.

Planned lead time

An overlapping of operations can either be specified using a send ahead quantity or lead time. The lead time describes the offset from the previous operation to its subsequent operation. A lead time can also be negative, if, for example, the subsequent operation begins with a setup before the previous operation.

Max. sync. time

If synchronization is activated with the subsequent operation by the [processing code](#) (customized), then it must be assured in the planning stage that the maximum time specified between this operation and the subsequent operation is the defined synchronization time

The time is calculated in hours based on the shift calendar.

Synchronization can be combined with an overlapping.

Planned wait time / wait time formula / minimum wait time

The wait time is one available option to absorb interferences and delays for each operation. It describes the (calculated) length of time that needs to pass before an operation can commence (setup) and it is accounted for during scheduling. The wait time can either be entered directly or it can be calculated using a formula.

The wait time can be reduced during the scheduling process. This requires that a reduction strategy has been entered for the order, which triggers a reduction in the wait time. The maximum amount of time it can be reduced is the minimum wait time.

Planned idle time

The planned idle time describes the length of time that needs to pass for processing related reasons before a manufactured or processed material can undergo the next processing step. The idle time is accounted for during scheduling. It is not possible to reduce the idle time.

Target transport time

The planned transport time is the time necessary to transport material from one workplace to the next. It is transferred from the upper-level ERP system or it can be calculated in MES using a transport matrix.

The transport time is accounted for during lead time scheduling and can also be reduced during this operation. The transport matrix and the reduction strategies are both defined when customizing the system.

Minimum transport time

The transport time can be reduced to this minimum amount of time during scheduling by applying reduction strategies.

The following wage specifications are used to calculate an incentive wage.

Wage type

Wage type

Wage indicator

Piecework ID/premium (E/G/S/M/Z/...)

Target te

Premium default: te (per 1,000 pieces).

"te" is the "time per unit" for each person. It is used to calculate the "order time", which is the specified processing time for each person used to calculate the incentive wage. The standard setting in MES is to show it in hours per 1,000 pieces and to transfer it at the interface in seconds per 1,000 pieces.

If no incentive wage is to be calculated in MES, you can enter "0" here.

Target tr

Target tr is the specified setup time (in hours) per person.

If no incentive wage is to be calculated in MES, you can enter "0" here.

Target teb

The premium default teb is the available machine time per unit and it is used to calculate the "occupancy time" for the workplace / the machine so that the incentive wage can be calculated. The standard setting in MES is to show it in hours per 1,000 pieces and to transfer it at the interface in seconds per 1,000 pieces.

If no incentive wage is to be calculated in MES, you can enter "0" here.

Target trb

Target trb is the specified setup time (in hours) per workplace / machine

If no incentive wage is to be calculated in MES, you can enter "0" here.

Processing tab**Processing code**

A processing code is a compilation of indicators that are used to control the operations. Each operation references this kind of processing code, and as a result its performance is defined in relation to the issues listed below.

This processing code is defined at the time the system is customized. Unless defined otherwise, the default setting to be entered here is the [processing code](#) SYSTEM.

Recordable

If this option is set, the operation can generally be logged in, provided other criteria is also met (e.g. operation not locked or operation cannot be logged in because of the status of a previous operation).

Can be logged on at the same time

This option specifies whether an operation may be logged on several times, i.e. at the same time.

This option should be enabled for overhead cost operations as well as for operations that are logged on to group workplaces. However, this option should not be set for operations that are subject to batch management requirement.

The ability to log on operations in parallel is not supported in the planning functions graphic detailed planning, order sequencing, graphic order sequencing. These planning functions assume that an operation is planned for exactly one capacity. If one operation is now logged on to several workplaces at the same time, contrary to capacity planning, this is then in opposition to planning. In order to conduct parallel planning of operations on different capacities, MES provides the operation splitting function.

Batch management requirement

Set this option if the operation requires batch management. This special processing step for operations that require batch management requires that material management is applied.

Serial number obligation

This option should be modified only after consultation with MPDV.

Layout

The code entered here references a label that was created in MES Label Designer that needs to be printed for the operation.

Target cycle

The target cycle is a work-operation related specification used for machine clocking. The target cycle does not depend on the number of produced parts. In MES, it is calculated and processed as a duration per 1,000 machine cycles.

If cycle time monitoring is active, this value is assumed as the default setting for finishing the operation. This value is the default value for the MDE machine monitoring function (cycle monitoring).

Partitioning

The partitioning (cavity) defines how many parts are produced during a machine cycle.

Partitioning is determined for each operation separately and is transferred via the ERP interface to MES. It is transmitted to the terminal at the time the operation is logged on and applies for the machine at which the operation is logged on.

Pulse factor

During automatic quantity collection at the terminal, a pulse factor stored at an operation is also considered. Consequently, the pulse factor and the partitioning represent a conversion factor for the automatic collection of quantities: $\text{primary quantity} = \text{cycle} * \text{partitioning/pulse factor}$.

Split authorization

This option defines whether an operation may be split.

Max. no. of splits

If an operation can be split it is checked whether or not the split number entered by the user exceeds the value entered here. If this is the case the split is rejected with an error.

M/O relation setup (machine/operator relation, setup)

Personnel requirements PEP (workforce requirements planning) for setting up the operation

Qualification (setup)

Unique qualification number from the qualification master data.

M/O relation production (machine/operator relation, production)

Number of employees required for production. By configuring the system accordingly during the customizing stage, you can define for each order type that only the number of persons specified here can log on. If several operations are logged on at the same time (in parallel), the maximum number of persons is equal to the total number of M/O relations for each separate operation.

In workforce requirements planning (PEP) this field is used to define the personnel requirements needed to produce the operation.



Alternatively to defining the personnel requirement by way of the machine-operator ratio for the operation, it can also be defined by way of required production resources. As opposed to the production resources, the personnel requirement for one qualification only can be defined for setup and production if the M/O ratio is used.

The machine/operator ratio (for setup and production) is then only relevant for personnel scheduling if a qualification has been entered in the field behind it.

Qualification (production)

Unique qualification number from the qualification master data.

Production variant

The use of the production variant when transferring order specifications from an external ERP system provides the option to define at which machine an operation should be planned. Furthermore, when making use of the graphic detailed planning based on existing production variants, one detailed plan can be carried out that accounts for the separate specification times (target cycle, setup and dismantling time) per production variant.

The key of the currently assigned production variant is filed here.

Data identifier

This is where the data identifier needs to be set if an Arburg Control System (ALS) is used. It must be unique (key). If ALS is not used, leave this field empty.

CBM tab

This index tab is only relevant in connection with the reel-based solution using in the material management module.

General

Special indicators

Not used; this field must remain empty.

Number of reels

The planned total number of reels to be produced (parent roll and sub-rolls); no specific processing in MES.

Material properties

Input width

Reel input width in MM

Output width

Reel output width in MM

If multiple rolls are manufactured at the same time in one operation, this field indicates the sum total of the separate widths.

If branches are planned, for each operation ("parent" and "sub-roll" operations), the output width of the separate operations is set explicitly (no sum total is generated).

Seam width

Total seam width in mm

If several reels are produced at the same time in one operation, this field will contain the sum total of the separate seam widths.

If branches are planned, for each operation ("parent" and "sub-roll" operations), the seam width of the separate operations is set explicitly (no sum total is generated).

Surface per piece

Surface for a piece in MM2/pce

Mass per unit area

Mass per unit area in G/MM2

Casing weight

This is where the casing weight for the sub-rolls is defined during cutting operations. Unit: G

Cutting information**Cutting OP**

Only relevant if the operation is a cutting operation

" " No roll cutting

"T" Roll cutting active (sub-roll numbering)

"M" Cutting active (parent rolls are being produced again)

Branch OP

Identifies parent and subordinate operations for a planned branch.

"M" Mother OP of a planned branch

"K" Child (subordinate) OP of a planned branch

Mother OP

If a branch is planned, the branched off material is allocated by these fields to the relevant mother operation.

A mother operation must reference itself.

Please note: The MES order ID (= MES order number = combined order / operation number) must be set here.

Daughter rolls/cut

Number of planned daughter reels per cut.

If the cutting plan is not planned, 0 is entered here.

Daughter rolls/cut - total

For cutting operations (mother OP): number of planned daughter rolls per cut (encompassing all branched off material).

If the cutting plan is not planned, 0 is entered here.

User fields tab

User fields offer the possibility to store further customer-specific information to MES besides the available fields in MOC standard. The operation-related user fields are displayed in the order information. The user field index tab is shown here for the operations, which, in addition to the user field key, also displays the defined user fields and includes a name and unit of measure. There are eight sub-index tabs in the index tab, which each have eight additional user fields. The so-called user field key determines which user fields are involved and which meaning they have.

Object type

The object type for the operation related user fields is AGNR (cannot be modified).

User field key

Every user field key describes a combination of user fields. The management of the user field key (and therefore the purpose of the fields) varies from one object to the next. User field keys are defined in coordination with the customer during the customizing process.

User fields

The following user fields are available after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

A maximum of 8 fields are shown for each page.

Default values tab

Up to ten default values can be entered for each operation. The default values are used, among other things, to calculate certain processing times using specified calculation rules. The meaning of each separate default value is defined by the default value key.

Please note

We recommend that you do **not** make any change to the default value key at the operation directly, because this might distort the meaning of the separate default values.

Default value keys are configured in coordination with the customer during the customizing process.

Administration tab

Created by / Created on

User who entered the operation as well as the time that the operation was entered. These fields cannot be modified.

Modified by / Modified on

User who most recently modified the operation, as well as the time that this modification was made. These fields cannot be modified.

Transferred by / Transfer time

Here, you can enter the source from where the operation was transferred. If the operation was transferred from the PPS system (PPS=J), the transfer time and date is automatically set to the time and date on which it was stored in MES. These fields cannot be modified.

Modified HYDRA

Information that states that the operation was changed in MES. This identifier is automatically set at "J", if a change is made in MES. This field cannot be modified.

Modified PPS

Information that states that the production order was changed in the ERP system. This identifier is automatically set at "J", if a change is made via an ERP interface (PPS=J). This field cannot be modified.

Deletion flag

This option is only displayed in the order information. It cannot be changed.

Responsibility area

If an area of responsibility is entered here, the user must have been authorized to display and edit operations and/or work plan operations.

The fields listed below are only displayed in the [order information](#). They cannot be modified by the user.

Locked / Locked by / Locked on

Locked operations cannot be logged in and are also not displayed on the terminal in the sequencing list - irrespective of how the status is configured.

Furthermore, the user is shown who was the last to lock the operation and also the time and date on which the last lock was made. These values remain even after the operation is unlocked. They are updated each time the operation is locked again.

Unlocked by / unlocked on

The user who terminated the last lock is shown here as well as the time and date on which the last unlocking occurred. These values remain even if the operation is locked again any time in the future. They are not updated until the operation has again been unlocked.

Locked for editing / Locked for editing by / Locked for editing on

Reserved; currently not used.

Reactivated by / Reactivated on

If an operation that has already ended is reactivated, the user is displayed here, who performed the last reactivation as well as the time and date on which the reactivation took place

24 Datenstruktur Arbeitsganglangtexte

Each of the fields for an operation long text are described below. The actual sequence of the editing dialogs may deviate from the one illustrated here.

MES order number / MES work plan number

Combined order/ operation number and/ or work plan/ operation number of the operation for which a long text is defined.

Short text

Short version of the long text, which is shown in the application list.

Long text

Actual long text, which is not shown in the application list.

The text entry function, which for the most part is equivalent to the functions of a text editor (highlighting of text passages; deleting or inserting of lines of text, as well as the merging of lines of text; copying with the key combination Ctrl+C, cutting with the key combination Ctrl+X, and pasting with the key combination Ctrl+V). Lines may have more than 80 characters when entered. When a document is saved, however, the system inserts a hard line break after the 80th character.

25 Data Structure of Components

Below is a description of the separate fields used for a (material) component. The actual sequence of the editing dialogs may deviate from the one illustrated here.

MES order number / MES work plan number

Combined order / operation number or work plan / operation number for the operation that a component has been assigned to.

Material

Enter the material number for the material component here.

Designation

Here you can specify the name of the material.

Comment 1 / Comment 2

These are comment fields.

BOM item

In a BOM, the separate components of a product are referred to as items. The order of the BOM item is determined by the number that is entered here. As a result, the same material numbers can appear in the component list and the entry can clearly be assigned to the correct component item.



Please note that when using the MPL, the BOM item must be **unique** within the operation.

For the reel-based solution "RF", this is the position of the component in the layer structure.

BOM level

A component can also have several levels. If applicable and known, please enter the BOM level here.



Postings can only be carried out on materials of a BOM level 0. If the system indicates a BOM level > 0, the component type (see the field after the next) must as a rule be set to "I" (info component).

Material type

Type of material of the material component. In MES, material specific processing is controlled by the material type.

Unless defined otherwise for a specific project, assign the material type SYSTEM here.



The material type must exist in MES (see configuration [material types](#)). If no material type has been entered, MES will attempt to determine the material component (requirement: the [assignment of material to material type](#) has been entered. If no material type can be found, MES will assign SYSTEM as the material type.



For info components (material type "I") we recommend that you define and assign a separate material type (e.g. INFO).

Component type

Possible values:

- | | |
|----------|---|
| M | Material component (default)
Generally, "M" should be entered here. Other component types may also be relevant for material management. |
| I | Info component
Info components make it possible to be shown in the component list without having to post them. |
| T | Carrier material (RF)
A maximum of one input batch may be logged into the machine as carrier material (T) or additional material (Z) at any one time. |
| A | Scrap/waste material (RF) |
| Z | Additional material as an alternative for carriers (RF).
A maximum of one input batch may be logged into the machine as carrier material (T) or additional material (Z) at any one time. |

Consumption type

The following options are available to enter material components. How the separate options and their application are defined depends, among other things, on which of the MES components / areas are used.

K = None

This option defines that no consumption is recorded for the material component. If this is the case, the material component will merely be displayed.

It is imperative that the info components (see above: component type) are set to this option.

L = Retrograde/ with batch reference (MPL/TRT, MPL-RF)

Choosing this option sets the system so the material component is logged in and out as a batch. The consumption calculation for these material components (retrograde, at input batch logout) depends on the configuration of the [material type](#) that the material component is assigned to.

D = Discrete



This option is relevant for *discrete consumption recording* (AIP-DVE). This type of material consumption recording needs to be configured especially while HYDRA is customized.

The option "L" needs to be used if Material and Production Logistic (MPL) or Tracking & Tracing (TRT) is used.

For this component, the system calculates consumption in reverse, which means the calculation is based on the last produced quantity and suggested in a posting dialog. Consumption is posted at the component level and the system generates a material movement (goods issue from production), which can be reported back to the upper-level ERP system.

Replaceable

If this identifier (=J) is set for this kind of component, you can choose to use a different material other than what was planned. However, only material of the same material type may be used.

Alternating (subject to change)

This option causes an input batch for a batch of this material to change to an output batch. The setting allowed for this option depends on the component type (see above):

Component type	Allowed setting options
M	☐ or ☒ possible
T, Z	Only ☒ allowed
I, A	Only ☐ allowed

Input quantity

Planned input quantity of the component per unit of the primary quantity of the operation.

Unit

Quantity unit for the input quantity

Input quantity in percent / upper tolerance limit / lower tolerance limit

Default: 0; should be modified only after consultation with MPDV.

Required quantity

The system computes the requirement quantity at display time. The calculation is based on the following formula: 'Requirement quantity = input quantity * target quantity of the OP in primary quantity unit.

Resource type

UOM Spec. mass per unit area

Spec. mass per unit area

Planned article

Backflush

Requirement quantity (PPS)

The "requirements quantity" field visualizes the total requirements quantity transferred from the ERP/PPS system, thus the required quantity of the component needed for the production of the operation quantity.

Consumption (total)

The "consumption (total)" column shows the total consumption that has been posted onto the respective component. In this context, it does not matter whether the component is subject to batch management or discrete.

Upper-level component: BOM item / BOM level

Reserved.

Modified by / Modified on

Editor as well as the date and time the last modification was made.

User fields

User fields can be defined and used for each specific project.

26 Production Resources & Tools Data Structure

Each of the fields for a production resource or tool are described below. The actual sequence of the editing dialogs may deviate from the one illustrated here.

MES order number/ MES work plan number

Combined order/operation number and/or work plan/operation number of the operation for which a production resource is defined.

Resource type

Resource type of the production resource or tool that is to be assigned to the operation. The resource type must be known in MES. Predefined resource types must be chosen from the selection menu. Additional resource types can be defined when customizing the system.

For documents, the resource type to be entered must be DOC.

Resource

Enter the resource number (material number) of the production resource.

Designation

Here, you can enter a name for the production resource or tool.

Comment 1/ Comment 2

These are comment fields.

Required quantity/ unit

Resource quantity required to carry out the operation. When planning the operation in graphic detailed scheduling, this number of resources is entered in terms of capacities.

The quantity unit is only used as a comment.

Path

When identifying a document as a production resource, the local reference to the path is to be defined in the [Path Configuration](#).

No path must be stored for DNC resources; it is determined based on the path stored for the resource type.

The field should be left empty for all other production resources.

File

When identifying a document as a production resource, the local reference to the path is to be defined in the [Path Configuration](#).

No file name must be stored for DNC resources; it is determined based on the path stored for the resource type.

The field should be left empty for all other production resources.



If a new document is assigned to an operation, it must be ensured that it exists at the stated location. No file is uploaded when a document is assigned!

Modified by/ Modified on

Editor as well as the date and time the last modification was made.